

Lead 101 and Tenant Organizing: We Keep Us Safe!

Science for the People Montreal, pamphlet presented at Constellation Anarchist Festival on May 17th 2025

KEY POINTS

- Lead is a toxic metal that can cause **long-term harm** at all stages of life; **no safe level of lead** or threshold in the human body has been identified.
- Hazardous levels of lead are widespread in Montréal. **Low-income tenants** living in **older buildings** are more likely to be heavily exposed, but everybody has some risk.
- Lead affects all organ systems in the body and can worsen an individual's health even in cases where symptoms are not obvious. Low levels of lead exposure are linked to cognitive deficits and behavioral problems in children, and an increase in premature birth and heart attacks in adults.
- Most household exposure occurs through ingestion or inhalation from lead-based **paint** and contaminated **soil**, as well as ingestion from **drinking water** due to lead-containing pipes and plumbing
- **The harms posed by household lead exposure are entirely preventable if tenants take collective action.** Tenants and tenant organizations can reduce lead exposure by educating themselves and their neighbours on the risks and pathways of lead exposure, demanding point-of-use filters with NSF/ANSI 53 (along with claims of lead reduction) and NSF/ANSI 42 certifications by default, and including professional lead abatement (in paint, soil, and intake pipes) as part of their list of housing demands.

What is lead?

Lead is a toxic metal with no known useful role in human biology. Children and pregnant individuals are among the most vulnerable to its toxic effects [1]. Lead continues to be present in many households and industrial products despite its known toxicity and despite the availability of non-toxic alternatives.

Where does lead go in the body?

Lead atoms can have a similar size and charge as calcium and zinc, allowing them to replace these essential elements and disrupt the functioning of important biological systems [1, 2]. After lead particles are ingested or inhaled, they quickly enter blood circulation, then gradually move into bones and organs (for a review, [3]). Different factors can cause lead stored in tissues and bone to be remobilized into the bloodstream throughout one's lifetime.

Why are children, pregnant people, and the elderly particularly susceptible to lead?

Children are at increased risk of ingesting lead in the environment; toddlers consume more dust than adults [4]. A child's digestive system absorbs lead more easily than that of an adult, and the developing brain and body is more vulnerable to its harmful effects [5]. Malnourishment exacerbates the effects of lead exposure – children absorb more lead if they are deficient in nutrients like calcium and iron [1]. Fetuses are exposed to lead during pregnancy, and newborns can be exposed to lead through breast milk [5].

Lead stored in the bones can re-enter the bloodstream as older adults age and experience bone loss. The high calcium demands of menopause and pregnancy cause increased bone resorption, which can cause lead to be suddenly released into the blood [6, 7].

A sliding scale of harms

While governments set thresholds to guide clinical or public health action, no safe level of lead exposure exists. (*These thresholds have different definitions, but tend to stand as a cutoff above which blood lead levels are deemed unsafe.*). The CDC uses a blood lead reference level of 3.5 µg/dL (in children) [8], while the Canadian blood lead intervention level is 10 µg/dL [9]. However, cognitive deficits have been shown to occur down to blood levels of 1 µg/dL [10]. Low levels of exposure during pregnancy have been shown to contribute to the chance of miscarriage [11], and an increased risk of developmental delays has been associated with levels as low as 1.67 µg/dL in umbilical cord blood [12]. Even when symptoms are not obvious, exposure to small amounts of lead can permanently harm a child's brain development and alter their ability to regulate attention and emotion. Beyond the nervous system, lead has been associated with health effects in every organ system, and the effects of exposure can be worsened by underlying or chronic health conditions (e.g. anemia, high blood pressure) [5].

If you are concerned about lead exposure, you can request a blood test from your doctor. Blood lead tests tell you about your recent lead exposure (e.g. the last few weeks), but these tests do not reflect past, long-term exposure, and may miss high-intensity brief exposures, both of which can impact your health.

How are tenants exposed to lead in their homes?

Landlords rarely inspect buildings while renting them and face little pressure to ensure safe living conditions.

In paint: If your dwelling was built before the 1960s, it likely contains lead paint. 31% of dwellings tested in 4 “older boroughs” of Montréal (Verdun, Saint-Laurent, Mercier-Hochelaga-Maisonneuve, Villeray-St-Michel-Parc-Extension) contained lead paint that exceeded US guidelines [13].

- **Sources:** Lead was a common additive in “premium” household paints sold throughout most of the 20th century [14]. Lead content was not limited in interior paints in Canada until 1976, when it was capped at 5% [15]. This cap was later reduced in 2005 to 0.06% and limited to “background levels” in 2010 (0.009%). Lead additives in exterior paint were not limited in Canada until 2005 [15]. If your dwelling was built before 1990, the exterior may contain lead paint [16].
- **Testing:** The City of Montréal does not offer paint testing. DIY at-home testing kits are relatively accessible but should be interpreted carefully: be aware of the possibility of both false positives and false negatives, and do not assume that a negative test means your household is free of lead contamination.
 - *Sodium rhodizonate* swabs work by indicating the presence of lead by turning red, pink or purple. LeadCheck brand swabs detect 0.5% lead in paint [17] but are difficult to acquire outside the US. Other swabs are available online, though their sensitivities have not been validated. We have provided Scitus swabs (grey, which must be activated with vinegar before testing, turning them mustard color) and non-branded swabs (mustard-coloured, must be activated with water before testing).
 - * Caution: False positives are possible. The mustard-coloured swabs can be triggered by copper and zinc, as well as lead. The Scitus ones are less prone to false positives, but are still not considered highly reliable.
 - * Advantages: Cheap!
 - * Disadvantages: Not highly reliable, not thoroughly validated, time-consuming to perform testing.
 - *Methylammonium bromide* sprays/ solutions bond with lead to form a mineral (perovskite) that glows green under UV light [18]. In this workshop, we are using the Lumetallix kit.
 - * Caution: This test is not highly informative of lead concentrations – extremely small amounts of lead will cause the surface to glow green with the same intensity as very large amounts of lead, even though the health risk posed by exposure to these amounts is not the same. Always read the test kit manual thoroughly.
 - * Advantages: Incredibly sensitive and highly reliable for detecting the presence of lead in a range of solid materials! Fast and easy.

- * Disadvantages: Cannot be used for non-solids, and cannot detect lead that is not directly on the surface of an object. It is expensive (\$104 CAD for a kit).
- Technique: Check all paint layers! Cut a notch with a box cutter through all layers so the test solution/swab can get in there. Prioritize surfaces that are deteriorating.
- **How to reduce exposure:** Demanding professional lead abatement services is the safest way to remove lead-containing paint and soil. Once you know that lead paint is present, avoid disturbing the walls to minimize dust exposure. If lead paint is exposed, it can be painted over with encapsulating paints to temporarily minimize exposure. It is extremely dangerous to remove lead paint with heat guns or sanders. More detailed guidelines are available at <https://www.canada.ca/en/health-canada/services/home-safety/lead-based-paint.html> [16].

In drinking water: The problem of lead-containing water service lines (as well as in-dwelling pipes, lead solder, fixtures and fittings) is widespread in Montréal. A 2019 investigation estimated that 300,000 people in the city were at risk of lead exposure from water [19]. While the city is replacing some lead service lines, this process is slow and far from complete.

- **Sources:** Lead service lines were installed prior to 1980 in Montréal [20] Lead solder was also used in potable plumbing until 1990 (National Plumbing Code [21]). Lead has not yet been banned from pipe and plumbing fittings (i.e. brass connectors), but limits were put into place in 2013. (National Plumbing Code [21]). In addition to dissolved lead, lead particles from these plumbing lines can break off into the water supply and end up in one’s drinking water.
 - Every service line has a public portion for which the city is responsible and a private portion for which the landlord is responsible (Figure 1). Montréal began replacing public lead service lines in 2006 [22]. As of 2019, 90% of homeowners living in areas where the city had begun replacing lead pipes had not chosen to replace their private portion of the service line [23]. In 2021, a new city bylaw took effect that would allow the city to begin replacing the private portion of entry pipes at the owner’s expense [24], but this process has been slow and is far from complete.
 - The City of Montréal provides a map of some information about the lead content of the public section of the service lines (“Map of lead service lines Montréal”: <https://montreal.ca/en/services/map-lead-service-lines>); however, landlords are still not required to disclose whether the intake pipes that lead to your tap contain lead or lead solder.

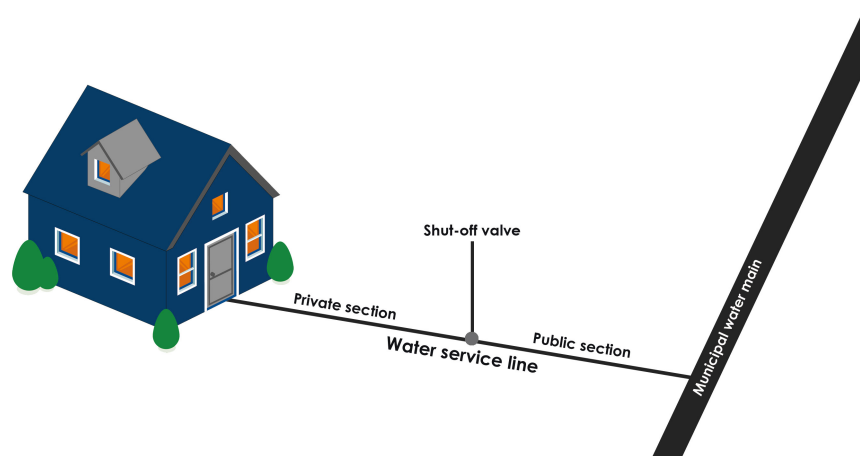


Figure 1: Different parts of the water service line are considered either the city or the landlord/ homeowner’s responsibility. Many landlords have not replaced their portion of the water service line and are not required to disclose whether the intake pipe contains lead. [25]

- **Testing:** The City of Montréal seems to currently perform on-site testing with the Palintest Kemio Heavy Metals sensor (disposable sensor anodic stripping voltammetry, DS-ASV; personal communication, 2025), but this testing program leaves many tenants falling through the cracks.
 - Standard lead in-water testing methods can miss worst-case lead levels [26, 27]. They can show ‘no lead’ or ‘low lead’ when lead is present and, at times, dispensed at very high levels. The

DS-ASV onsite testing method can only measure dissolved lead, often underestimating the total lead in water [28].

- To have the city test your water for lead, you can fill out the “Formulaire de demande de dépistage d’entrée d’eau en plomb” form if your building has less than 8 dwellings and was built before 1970. (<https://montreal.ca/demarches/demander-le-depistage-dune-entree-deau-en-plomb>)
- Note: the electronic indicators that sometimes accompany the ZeroWater filters provided by the city provide an indication of the total dissolved solids in the water (including salts, minerals, metals, and other substances), and are intended to be used to determine when to replace your filter, not as an indication of water lead content.
- **How to reduce exposure:** Use filters with NSF/ANSI 42 and NSF/ANSI 53 certifications, along with claims of lead reduction. Filters can be in a pitcher or attached directly to your tap. Make free certified filters and filter replacements part of your demands as a tenant. Filters must be replaced regularly to be effective.
 - As of 2019, you can request a one-time free ZeroWater filtration pitcher and a single set of filters by calling 311, if the government deems your household “eligible.”
 - Filtering is advisable even if you haven’t tested or the test came back indicating relatively low levels, since sources of particulate lead are often undetectable [28].
 - Avoid first-draw water: run the tap for several minutes or until it gets cold.

Other sources of exposure:

- **In contaminated soil:** Tetraethyl lead was added to gasoline before being limited in Canada in December 1990 [29]. This compound gives lead dust upon combustion, which comes out of the tailpipes and settles. Much of it remains in soil, attics, etc.
- **In household dust:** Lead in household dust usually has its source in contaminated soil and interior and exterior lead paint, but other sources of lead can contribute. Examples include: lead from batteries, remnants of previous renovations, pyrotechnics, wheel weights, battery cleaning brushes, ammunition, solder, pottery, hobby casting of fishing weights, etc.

1 Practical testing

Testing paint: The City of Montréal does not appear to offer paint testing. DIY at-home testing kits are relatively accessible. **But for reliable information, all tests must be used as part of a larger technique.** No test, not even the best with the most expensive and fancy equipment, will give you reliable answers all on its own.

- Sodium rhodizonate swabs work by indicating the presence of lead by turning red, pink or purple when the test solution absorbed in the tip of a swab reacts with the lead ions. Rhodizonate is a grey solid, but the solution is orange. The test solution has to have a low PH (be acidic) to dissolve lead from the solid surface/get the ions into solution so it can react with the rhodizonate.
 - If the color of the object you are testing is red or purple, clearly that could interfere with the test.
 - Rhodizonate based swabs can give false negatives if there is calcium present. The plaster in drywall is mostly calcium sulfate. So don’t get any plaster from the wall on the test area etc.
 - LeadCheck brand swabs detect lead in paint quite reliably (**Scharman & Krenzelok, 2008**) but are difficult to acquire outside the US. They contain 2 glass ampules, which contain the rhodizonate and an aqueous solution with an acid. The chemistry of the test solution is better controlled and specified, making them more reliable, repeatable, and sensitive. \$5 each, approximately.
 - Cheapie swabs
 - * The cheap mustard-coloured swabs
 - \$0.16 each

- False positives: can be triggered by copper or zinc as well as lead. They go purple or red, depending on the exact combination of metal ions present.
 - Activate them by dipping in water, get excess water off by pressing it gently against the side of the container with the water, then rub on the surface to be tested in accordance with the instructions.
 - They do seem to be pretty sensitive though.
 - It's not actually exactly clear how they work.
- * The grey Scitus ones
 - \$1 each
 - The rhodizonate powder is just mixed with the cotton on the swab tip. That's why it's grey.
 - are not apparently prone to false positives.
 - Activate them by dipping them in **vinegar**, not water. Get excess of, then apply to area according to instructions. You have to rub it for a while to get the lead to dissolve from the surface, as noted in the instructions.
 - Lead makes the swab more of a purple color than red.
- **Methylammonium bromide solutions.** When the solution is applied, a reaction occurs, basically instantly, at the surface of a solid containing lead to form perovskite(s), which glow green under 254 nm UV light. We have tried the Lumetallix kit and it's pretty good. There is another very similar one called Flouoro-spec which is probably also ok but we haven't tried it.
 - Incredibly sensitive and highly reliable for detecting the presence of lead in a range of solid materials! Fast and easy.
 - Expensive (\$104 USD)
 - Caution: This test is not highly informative of lead *concentration* – small amounts of lead will cause the surface to glow green almost as well as large amounts.
 - You can use it to play detective, identifying the source of contamination with greater ease than swabs allow.
 - Shine the light first to see if there is anything glowing green without the test solution. Sometimes there is some dust that glows anyway (which, in the absence of the test solution, would not indicate the presence of lead).
 - Very low false positive rate. Beware anything like grease, oil, or another layer of paint on the surface could prevent a green glow even if there is lead, because the test solution cannot react with the surface. If the UV light is blocked, same deal, though that doesn't seem to happen much.
 - It has to be dark enough you can see the green glow. It's pretty bright but you can't see it well in sunlight of course. It's also hard to get a good picture.
- **Technique:**
 - Check all paint layers! Cut a notch with a box cutter through all layers so the test solution/swab can get in there. "Alligator cracking" is common for old lead paint. It's the presence of a network of cracks all over the paint layer, often visible through later layers. You see it on doors in Montreal fairly often.
 - If you get a positive, it's best to confirm, with another swab and ideally another method. Try to figure out exactly where the lead is coming from.
 - If you get a negative with a swab, you can test another surface, but only within a minute or two after activation before the test solution doesn't work any more.
 - Check the most troublesome areas first. Sliding windows, anywhere the paint is abraded or deteriorating.

XRF and lab testing

- XRF stands for X-ray Fluorescence. It is implemented using an amazing but very expensive handheld instrument. Some cities have people running around doing analysis with the instrument all the time, making it economical. They are about \$30,000 USD each, and require a license to use/own due to the small amount of radioactive material in them, which emits the x-rays.

- You still need to use them with technique, and interpret the results effectively. They need to be calibrated, for instance.
- The obvious problem is you have to trust the operator, which in this society is a big ask. Not only to tell the truth, but also to do a good job.
- You can collect paint samples and send them to a laboratory, which usually uses mass spectrometry or optical spectrometry. There are three different instruments that are commonly used, but it doesn't really matter much to the customer which one, they all work well within the box of what they do. They aren't necessarily universally better than swabs or Lumetallix though because you only get one test, it takes a long time to get the results, and you need to trust the third party. This may still leave you uncertain about other areas of the home, where it comes from or whatever. They could be good for final confirmation of a positive that could impress politicians or whatever that don't understand how such testing works.

Testing dust and soil:

- The swabs and Lumetallix also work for dust, but only in relatively high concentrations. You can get test kits for dust where you clean a square foot of area and then send the cloth to a lab to test for lead.
- Soil is harder to test for as concentrations below the limits of the swabs, and the practical realities of the Lumetallix, can still be a problem. XRF units can analyze soil but only in bulk, not the material on a surface.

Testing random consumer stuff

- Plastic (e.g. vinyl is particularly bad as lead is deliberately added as a stabilizer sometimes): no special precautions
- Pottery: swabs can have a hard time getting the lead detected as it can be bound tightly to the ceramic materials. Lumetallix works well for this.
- Solder: all three swabs work well, but remember copper and zinc are often nearby and can mess up the cheapie swabs.

Testing blood

- Testing the actual bloodstream is a good trick but not really the first line of defense, unless lead exposure is suspected. Most people most of the time probably shouldn't bother.
- You get a blood sample drawn, and the sample is analysed using the same kind of lab instrumentation used to analyze dust and paint samples, mass spectrometers or optical spectrometers.
- Great way to measure actual exposure. The results from the environment are only a proxy for how much is actually getting ingested or inhaled, and keeping them low is part of the safety margin. But if Blood Lead Levels are high, it's a problem already.
- Usually measured in micrograms per deciliter (tenths of a liter).
- Any primary care provider can order the test and it's not expensive compared with other blood tests.
- If elevated levels are detected, it makes sense to test repeatedly to ensure that mitigation efforts have succeeded.
- Remember it takes a while for the levels to go down after you stop ingesting/inhaling lead. The half-life of blood lead levels varies considerably between people/circumstances, but is often reckoned to be about 25 days.

Positive control

- It's useful to have a known lead contaminated object so you can be sure the test is working, or familiarize yourself with what a positive looks like. Obviously some of these are inherently dangerous because as explained the quantity of lead that it takes to cause harm is extremely small. Do not sand them or whatever.

- “Sweat joints” on copper supply side plumbing built before 1992 or so (lead solder). This is probably the easiest for many people. Be careful to avoid the copper, which as mentioned above can cause false positives.
 - * You can see what copper looks like when the test detects it. Also zinc, using any galvanized object.
 - * If you have some spare swabs or want to be really thorough, you can also deliberately get some plaster from gypsum wallboard on the swab, it should slowly turn black usually, for the 3M swabs (it takes a while, can be hours). Nothing seems to happen with the Scitus or cheapie swabs. You may need to put it in a ziploc plastic bag to prevent water from evaporating meantime.
- Lead solder. It’s still sold at some hardware stores, fuck them, some are better than that. Electronics made before 2006 sometimes used lead solder, before 2003 it is common.
- Lead wheel weights, common on cars
- Lead acid batteries. The terminals are often pure lead.
- Shot for shotguns, bullets, pellets for pellet guns may be kicking around.

If you are concerned about lead in your building and want to organize for repairs, contact The Montreal Autonomous Tenants’ Union. (Instagram: slam.matu; email: slam.matu@protonmail.com)

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